

TURAI, Saudi Arabia

WMO: 403560

Lat: 31.693N

Lon: 38.731E

Elev: 2803

StdP: 13.27

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.0	(c) 33.4	(d) 10.0	(e) 10.2	(f) 53.1	(g) 12.6	(h) 11.6	(i) 54.5	(j) 27.0	(k) 48.4	(l) 24.3	(m) 47.9	(n) 6.9	(o) 100	(p) 0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	27.4	105.7	65.8	102.5	64.9	100.4	63.8	68.1	95.2	66.9	94.2	65.8	93.0	8.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.0	88.9	73.4	59.0	82.8	72.7	57.1	77.1	72.5	34.0	94.3	33.0	94.2	32.1	91.7	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.2	21.0	18.5		DB	24.8	108.8	3.1	1.8	22.6	110.1	20.8	111.2	19.1	112.2	16.9	113.5
				WB	22.4	70.1	3.2	2.1	20.1	71.6	18.2	72.8	16.4	74.0	14.1	75.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.0	45.9	49.6	56.8	65.5	74.9	81.4	85.3	85.8	80.4	71.3	57.4	48.8
	DBStd	15.34	5.17	5.96	6.89	7.14	6.12	4.49	4.34	4.45	4.74	5.87	6.41	5.47
	HDD50	330	146	72	13	1	0	0	0	0	0	0	12	86
	HDD65	2135	591	433	269	81	5	0	0	0	0	17	237	502
	CDD50	6550	20	60	226	465	773	943	1095	1110	913	662	234	49
	CDD65	2879	0	1	16	95	313	493	630	645	463	213	10	0
	CDH74	40115	0	17	260	1193	3980	7011	9503	9622	6053	2328	144	4
	CDH80	23187	0	1	71	410	1946	4148	6047	6157	3434	953	20	0
(14) Wind	WSAvg	9.6	9.2	9.8	10.5	10.6	10.4	10.2	10.1	9.4	8.9	8.7	8.5	8.3
(15) Precipitation	PrecAvg	2.1	0.2	0.5	0.2	0.1	0.1	0.0	0.0	0.0	0.0	0.3	0.2	0.5
	PrecMax	5.4	1.1	1.5	1.4	0.7	1.1	0.1	0.0	0.0	0.4	3.7	1.4	1.5
	PrecMin	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.1	0.3	0.4	0.4	0.2	0.2	0.0	0.0	0.0	0.1	0.7	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	76.8	87.6	92.8	99.0	105.4	108.7	108.0	104.2	95.1	82.7	73.0
		MCWB	50.6	54.0	56.9	58.9	62.7	64.1	66.7	66.7	65.5	62.5	56.9	53.4
	2%	DB	64.3	70.2	80.9	88.2	96.4	101.8	105.6	105.9	100.5	91.6	78.5	67.8
		MCWB	49.0	51.0	54.6	57.4	61.1	63.7	65.8	66.0	64.0	60.7	55.9	51.0
	5%	DB	60.5	66.1	75.5	84.5	93.1	98.7	102.4	102.6	97.1	89.2	73.8	63.0
		MCWB	47.6	49.1	52.7	56.1	60.0	62.1	64.9	65.2	63.0	60.3	54.6	49.7
	10%	DB	56.9	62.3	71.6	81.0	89.8	96.5	100.0	100.3	94.7	85.9	70.1	59.4
		MCWB	46.4	47.5	51.4	55.0	58.7	61.4	63.7	64.2	62.5	59.8	53.6	48.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	56.6	60.1	62.4	64.9	67.2	69.2	70.1	68.3	66.0	61.3	56.9
		MCDB	62.9	70.3	76.2	81.2	91.6	95.0	97.6	95.6	93.2	87.3	72.4	65.5
	2%	WB	50.4	52.9	56.6	60.0	63.2	65.5	67.5	68.6	66.3	63.7	59.2	53.8
		MCDB	59.4	65.2	74.1	79.3	89.4	93.9	96.0	95.7	92.1	83.5	70.6	62.8
	5%	WB	48.6	50.7	54.3	58.0	61.6	63.9	66.5	67.3	65.1	62.2	57.2	51.5
		MCDB	57.8	62.4	72.6	79.6	87.2	92.4	95.2	94.2	90.6	82.1	68.4	60.2
	10%	WB	47.2	48.7	52.4	56.4	60.2	63.0	65.4	66.1	63.8	61.0	55.5	49.5
		MCDB	55.5	59.7	69.2	77.8	85.3	91.1	93.6	93.0	88.9	80.0	66.4	57.8
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	21.0	23.3	24.2	25.5	27.1	27.3	27.4	26.6	24.8	22.1	20.1
		MCDBR	25.1	27.6	28.6	28.0	27.9	29.4	29.0	28.7	28.3	27.7	25.9	24.7
		MCWBR	13.3	13.3	13.3	11.7	10.6	10.2	9.4	9.4	10.4	11.4	12.5	13.5
		MCDBR	21.4	23.3	25.9	24.9	25.4	27.3	27.6	27.8	26.5	24.1	21.1	21.1
	5% WB	MCWBR	12.4	12.4	12.4	11.1	9.9	9.4	8.9	9.2	9.8	10.1	10.8	12.7
(40) Clear-Sky Solar Irradiance	taub		0.329	0.357	0.406	0.475	0.479	0.407	0.402	0.409	0.396	0.426	0.367	0.337
	taud		2.269	2.174	2.015	1.841	1.857	2.076	2.116	2.097	2.115	2.015	2.158	2.245
	Ebn at Noon		280	283	276	261	260	279	279	276	275	254	262	270
	Edh at Noon		35	42	53	66	65	52	50	50	47	49	38	34
(44) All-Sky Solar Radiation	RadAvg		1109	1394	1804	2123	2394	2652	2551	2338	2023	1564	1206	1028
(45)	RadStd		51	90	67	68	104	41	59	50	65	70	77	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	+1.87	+2.30	+2.84	N/A	N/A	N/A	N/A
Regional (3 neighbors)	+0.65	N/A	N/A	+1.06	+1.82	+1.91	N/A	N/A	+225	N/A

Nomenclature: See separate page